SCOOT · Foundation · Design workshop

Designing SCOOT for a town, step by step

From 'is it worth it?' to switch-on day: seven steps an engineer follows to put adaptive signal control on a real urban network.

smrahmanbd.github.io/scoot-learning-lab

Seven design steps, each feeding the next

1 Decide whether SCOOT is worth it

~20% less delay than typical fixed-time plans (TAL 7/99)

2 Survey the network

1800 veh/h per lane: a typical saturation flow

3 Group the nodes into regions

60 | 90 seconds: Riverside's two region cycles

4 Define links and place the detectors

8–15 s ideal warning time from loop to stop line

5 Wire it up and build the database

144/s loop readings for Riverside's 36 loops

6 Validate every link on the street

0.59 LPU/s: N3A's validated SATO

7 Commission, fine-tune and evaluate

13–15% shorter peak journeys in Riverside (illustrative)

Three surprises to start

The software is the easy part.

SCOOT comes as a finished product. Most of an engineer's time goes on the street: counting traffic, choosing where loops go, and standing at stop lines with a stopwatch until the model matches reality.

The loops go 13 seconds before the lights.

An actuated signal puts loops at the stop line. A SCOOT loop sits near the upstream end of the road, about 8 to 15 seconds' drive away, so the model gets warning of every platoon.

A good fixed plan goes stale.

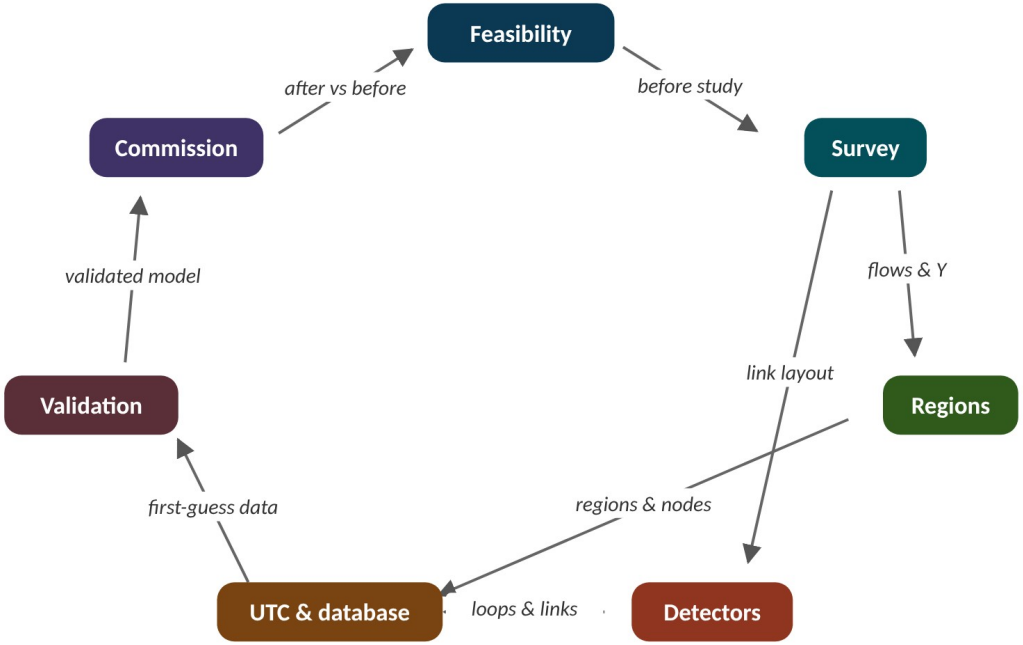
Traffic patterns drift, so a fixed-time plan loses about 3% of its quality each year (TAL 7/99). SCOOT re-times the network every few minutes instead of every few years.

Words you will meet

Node	A junction or pedestrian crossing controlled by SCOOT.
Region	A group of nodes that share one cycle time.
Link	One approach into a node, from its detector to the stop line.
Critical node	The busiest node in a region; it sets the region's cycle.
Double cycling	A quiet node runs two cycles in the region's one.
Loop	A wire coil cut into the road that 'feels' a vehicle above it.

Outstation	The box in the signal cabinet that talks to the centre.
UTC	Urban Traffic Control: the central computer system SCOOT runs on.
Saturation flow	Vehicles per hour that cross the stop line from a queue on green.
Validation	Tuning each link's model until it matches the street.
Commissioning	Switching SCOOT on and checking it works as designed.
Before/after study	Journey-time surveys that show what the scheme gained.

The map of today's ideas



The whole journey in one picture

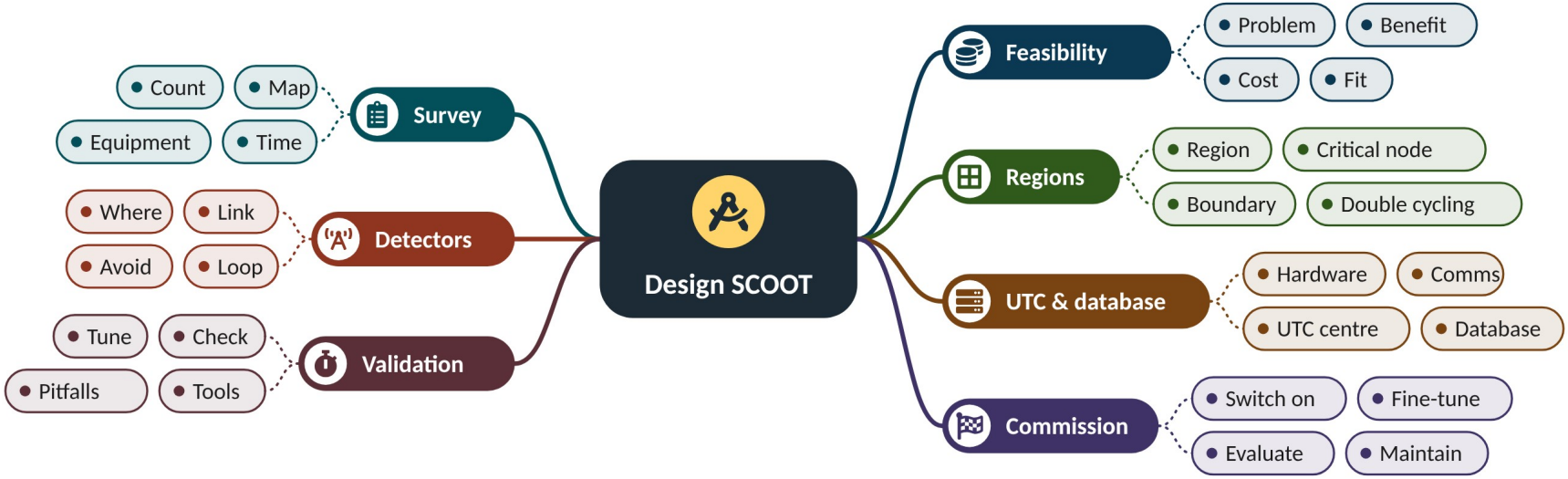


NOTICE 1 Every step produces something the next step needs: no step can be skipped.

NOTICE 2 Steps 2, 4 and 6 happen on the street; steps 3 and 5 happen at a desk.

NOTICE 3 Step 7 never really ends: loops fail and traffic changes, so the model is checked every year or two.

Mind map: the seven design steps



1 Decide whether SCOOT is worth it

IN PLAIN WORDS

Buying SCOOT is like fitting a smart thermostat to a big building. You first check how much energy you waste now, how much the gadget saves, and how long until the savings repay the price.

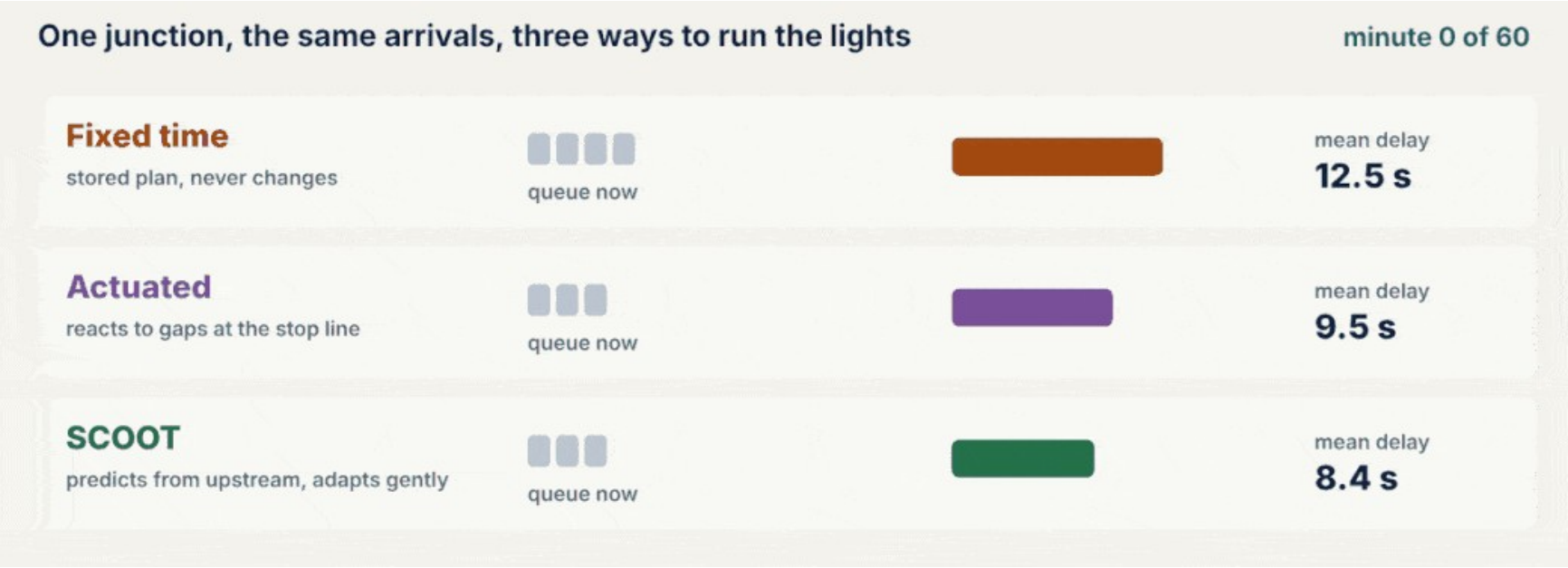
- Measure today's delay: the before study
- Expect about 12% (fresh plans) to 20% (typical plans) less delay
- Plans age about 3% a year
- Payback = cost ÷ yearly saving

~20%

less delay than typical fixed-time plans (TAL 7/99)

Source: Benefit figures and plan ageing: DfT Traffic Advisory Leaflet 7/99.

Watch: Fixed, actuated and SCOOT side by side



NOTICE 1 Fixed time cannot react: the plan was made for yesterday's traffic.

NOTICE 2 Actuated signals react, but each junction on its own.

NOTICE 3 SCOOT reacts and coordinates, which is where the delay saving comes from.

Worked example: Is Riverside worth it?

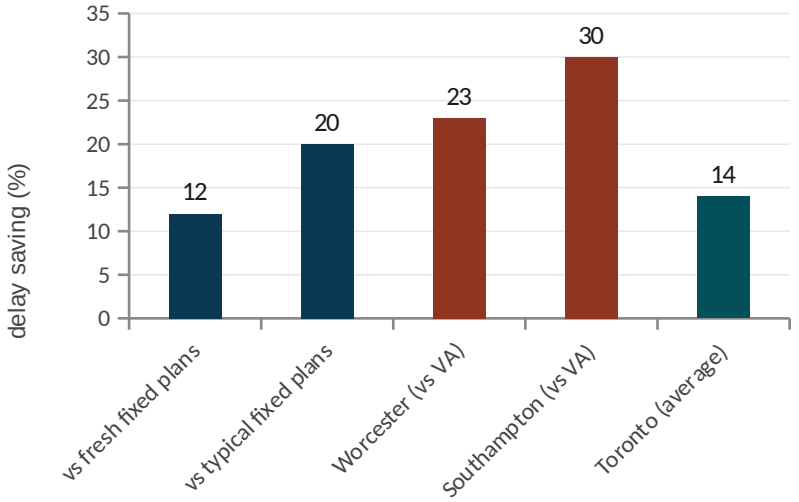
Yearly saving = daily delay × saving % × working days × value of time
Payback (years) = scheme cost ÷ yearly saving

Before study: 400 vehicle-hours of delay each weekday on old plans. Value of time 10 money units per vehicle-hour (illustrative). Scheme cost 400,000.

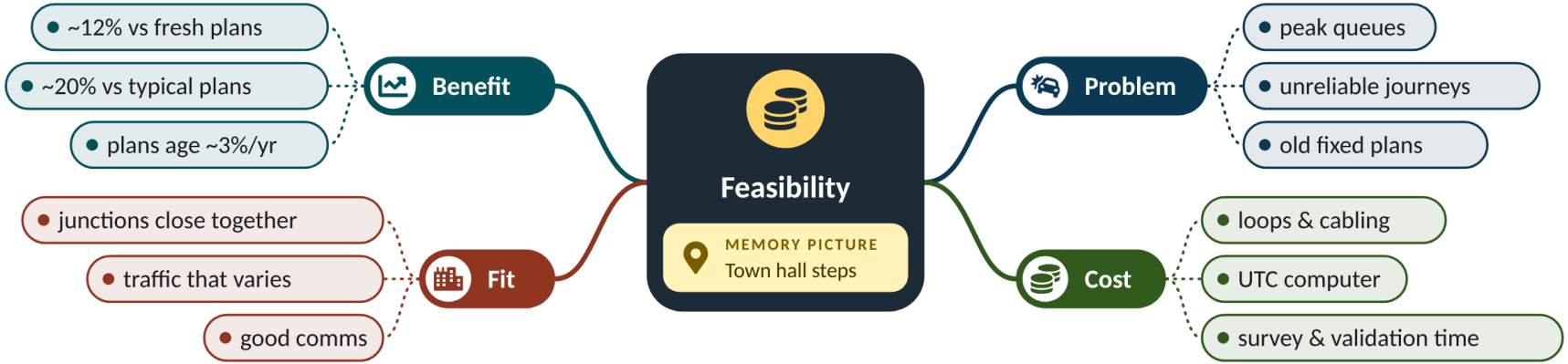
1. Expected saving vs typical plans ≈ 20%, so 400 × 0.20 = **80 veh·h per day**.
2. Over 250 working days: 80 × 250 = **20,000 veh·h per year**.
3. In money: 20,000 × 10 = **200,000 per year**.
4. Payback = 400,000 ÷ 200,000 = **2 years**.

Riverside repays the scheme in about 2 years, if the 20% saving is achieved.

Reported delay savings with SCOOT (TAL 7/99).



Remember it: Feasibility



PICTURE THIS · STOP 1: TOWN HALL STEPS

On the town hall steps sits a giant piggy bank with a 20% slice cut out like a cake. Every year the mayor must feed it 3 coins or it goes stale and smelly.

TRAP

The 12–20% are averages from past schemes, not a promise. A network with fresh, well-made plans may gain less; a badly timed one may gain more. Always measure with your own before study.

2 Survey the network

IN PLAIN WORDS

Before a tailor cuts cloth, they measure you: chest, arms, legs. The survey is the town's measuring session; SCOOT is then cut to fit.

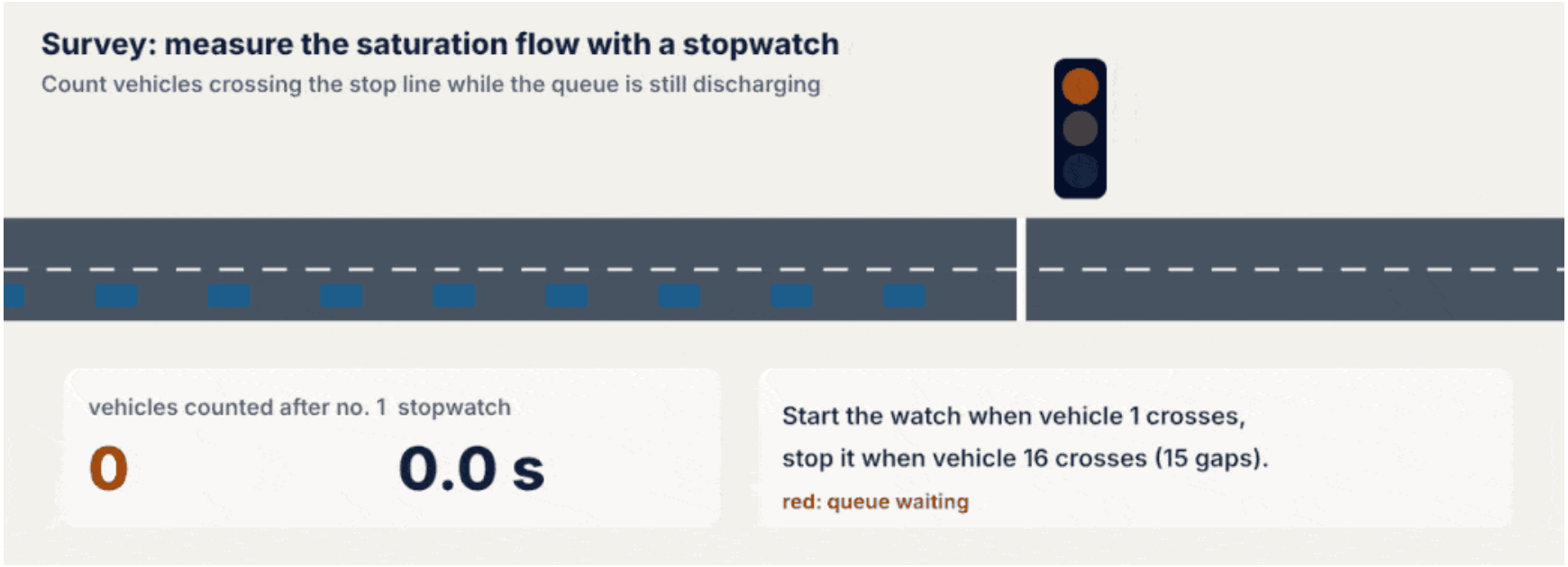
- Map every node: lanes, stages, controller
- Count turning flows, peak and off-peak
- Time journey times along each link
- Stopwatch the saturation flow

1800

veh/h per lane: a typical saturation flow

Source: Survey items: TRID 186642 (practical implementation of SCOOT); saturation flow method: standard practice (TSM Ch. 6).

Watch: Measuring the saturation flow



NOTICE 1 The watch starts when vehicle 1 crosses the stop line.

NOTICE 2 Vehicles 2 to 16 cross at a steady 2 s spacing while the queue lasts.

NOTICE 3 15 vehicles in 30 s gives 0.5 veh/s, or 1800 veh/h per lane.

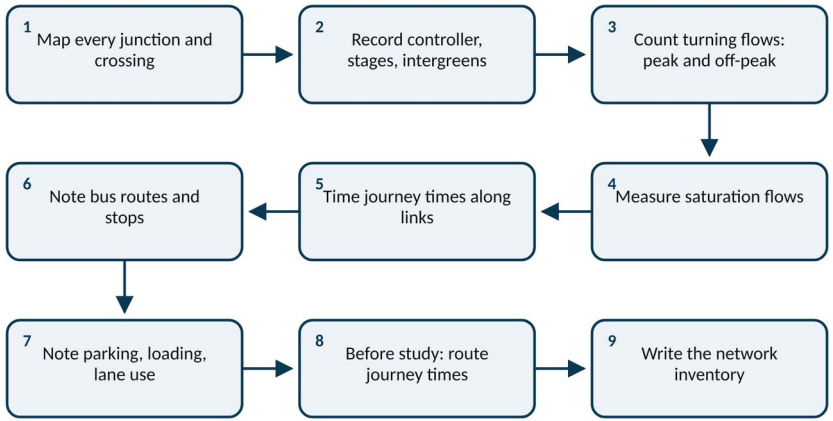
Worked example: Saturation flow and capacity at Riverside J3

s = vehicles counted ÷ time (start timing at vehicle 1)
capacity = s × green ÷ cycle; DoS x = flow ÷ capacity

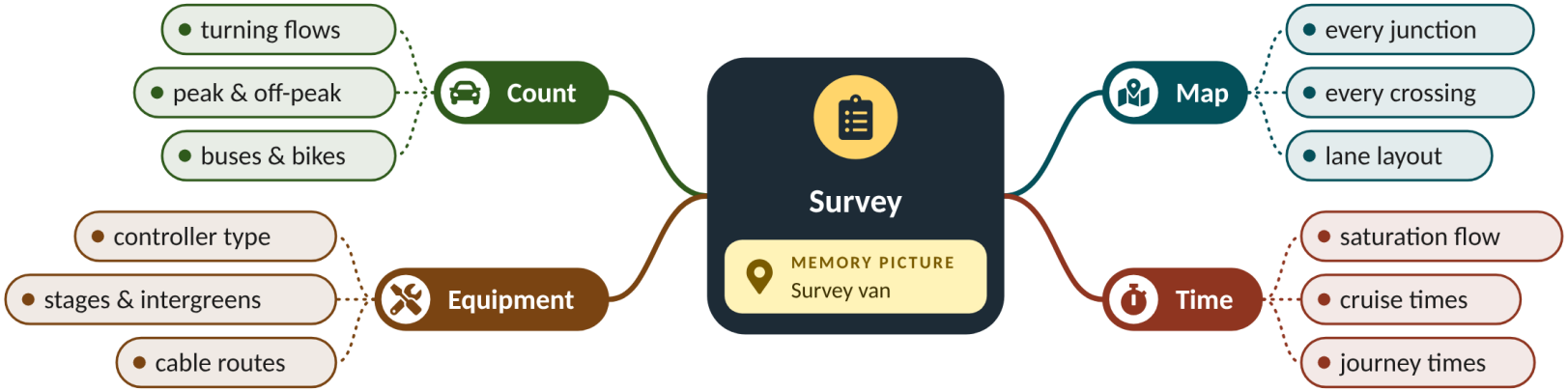
Two-lane main approach. On green, 15 vehicles cross in 30 s after the first one (per lane). Green 27 s in a 60 s cycle. Flow 1300 veh/h.

1. s = 15 ÷ 30 = **0.5 veh/s per lane** = 1800 veh/h per lane.
2. Two lanes: 2 × 1800 = **3600 veh/h**.
3. Capacity = 3600 × 27 ÷ 60 = **1620 veh/h**.
4. Degree of saturation x = 1300 ÷ 1620 = **0.80**.

The approach runs at 80% of capacity: busy, but not yet at SCOOT's 90% target.



Remember it: Survey



PICTURE THIS · STOP 2: SURVEY VAN

A survey van has a giant stopwatch for a headlight. When it clicks, 15 tiny cars hop out of its door in 30 ticks, and a clipboard on the roof writes '1800'.

TRAP

Do not count only the morning peak. SCOOT runs all day, so the survey needs off-peak and evening counts too, and it should note events that cause surges (schools, football, car parks).

3 Group the nodes into regions

IN PLAIN WORDS

A region is like a marching band: everyone keeps the same beat. The drummer is the critical node, the busiest junction, and the others march in step. A second band far down the road can keep its own beat.

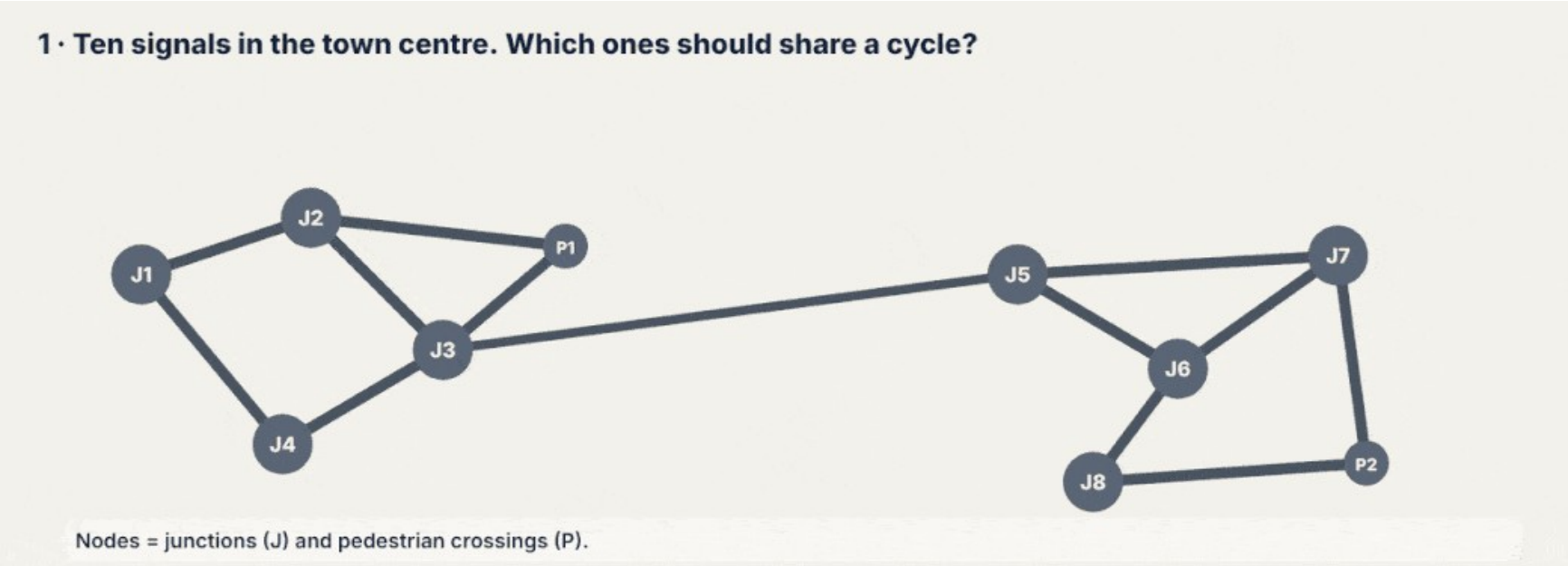
- A region shares one cycle time
- The critical node sets it, at about 90% saturation
- Split regions at long links
- Crossings and quiet nodes can double-cycle

60 | 90

seconds: Riverside's two region cycles

Source: Regions, common cycle, double cycling and the 90% target: DfT Traffic Advisory Leaflet 7/99.

Watch: Dividing Riverside into regions



NOTICE 1 Each node's needed cycle comes from its own traffic.

NOTICE 2 The gold-ringed critical node sets the cycle for its whole region.

NOTICE 3 The 900 m link is the natural boundary; P1 and P2 run twice per region cycle.

Worked example: Two regions for Riverside

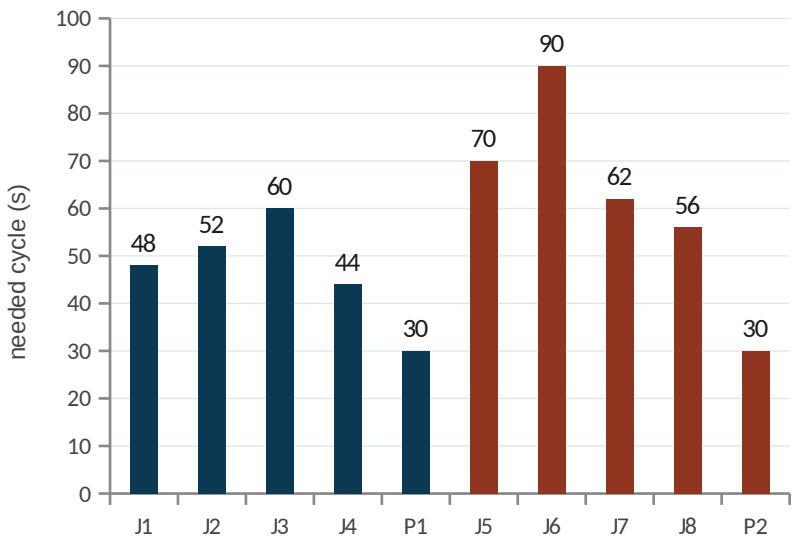
node DoS = Y × C ÷ (C − L) → needed cycle for 90%: C = 0.9 L ÷ (0.9 − Y)

Lost time L = 12 s at every junction. J3 has Y = 0.72; J6 has Y = 0.78. The link J3–J5 is 900 m long; cruise speed 13.3 m/s.

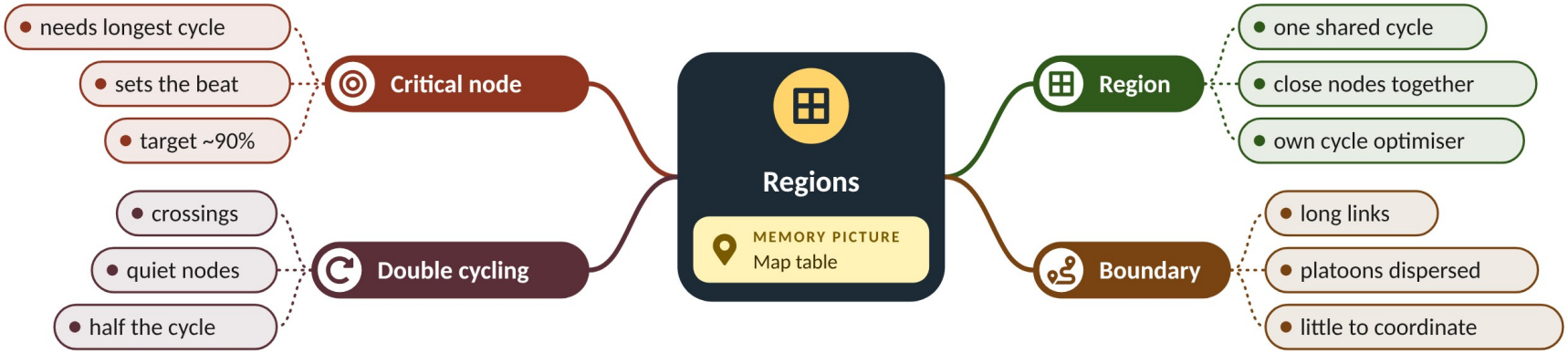
1. J3: C = 0.9 × 12 ÷ (0.9 − 0.72) = 10.8 ÷ 0.18 = **60 s**.
2. J6: C = 10.8 ÷ (0.9 − 0.78) = 10.8 ÷ 0.12 = **90 s**.
3. J3–J5 takes 900 ÷ 13.3 ≈ **68 s** to drive: a natural boundary.
4. Region A runs 60 s, region B 90 s. Crossing P1 needs only 30 s, so it double-cycles (2 × 30 = 60).

Two regions: A at 60 s set by J3, B at 90 s set by J6. One region at 90 s would make every red in A longer for no benefit.

Cycle each Riverside node needs for 90% saturation (s).



Remember it: Regions



PICTURE THIS · STOP 3: MAP TABLE

On a huge map table sit two jelly moulds full of junctions. In each mould a junction wearing a gold crown beats a drum, one at 60, the other at 90. A 900 m strip of liquorice keeps the moulds apart, and two tiny crossings tap twice per beat.

TRAP

The formula C = 0.9 L ÷ (0.9 − Y) is a classroom shortcut that shows the idea. SCOOT's cycle optimiser works on its own modelled degrees of saturation, in small steps every few minutes.

4 Define links and place the detectors

IN PLAIN WORDS

A loop is like a lookout posted down the road who shouts 'cars coming!'. Post them too close and the shout arrives with the cars; post them a little way up the road and the gatekeeper has time to get ready.

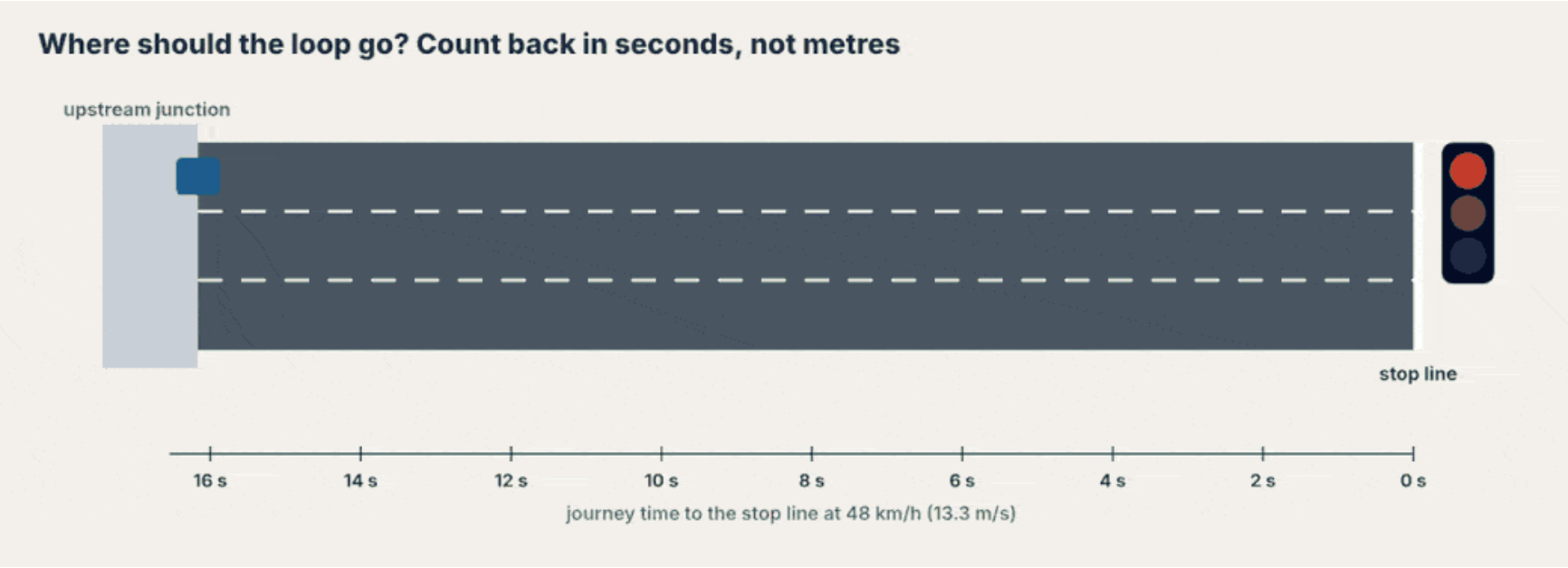
- One link = one approach, detector to stop line
- Loop at the upstream end: 8–15 s, at least 6 s
- 2 m loop, covering at most two lanes
- Keep the journey time consistent

8–15 s

ideal warning time from loop to stop line

Source: Loop size, lanes per loop, 6 s minimum and 8–15 s optimum: Davies (SCOOT design principles); one loop per two lanes: FDOT (2016); detectors on every link, upstream end: DfT Traffic Advisory Leaflet 7/99.

Watch: Counting back in seconds



NOTICE 1 The ruler under the road is in seconds of driving, not metres.

NOTICE 2 Red is too late (under 6 s); green is the ideal 8–15 s.

NOTICE 3 Each car flashes the loop as it passes: that count is SCOOT's input.

Worked example: Placing the loop on link N3A

distance = speed × journey time; 48 km/h = 13.3 m/s

6 s → 80 m 8 s → 107 m 15 s → 200 m

The link is 200 m from the upstream junction to J3's stop line. Speed 48 km/h (13.3 m/s). Three lanes.

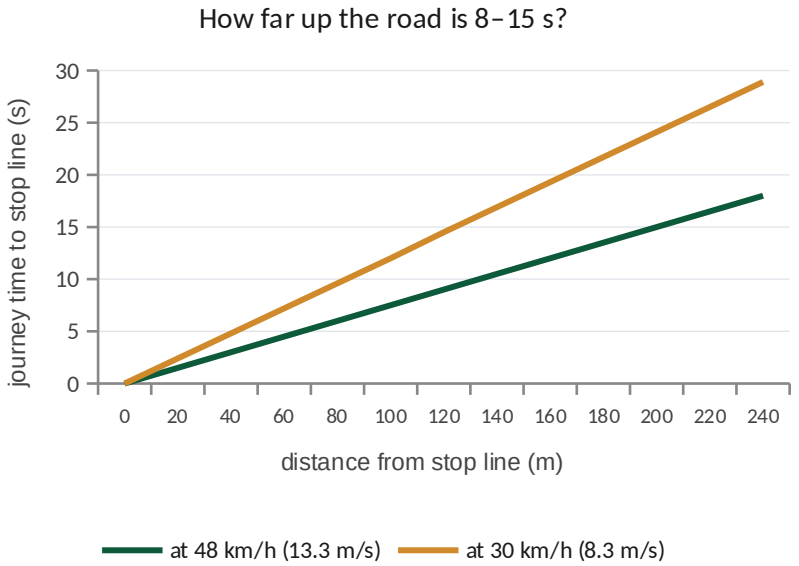
1. Ideal zone: 8 × 13.3 = **107 m** to 15 × 13.3 = **200 m** from the stop line.

2. Put the loop 20 m past the upstream exit: 200 − 20 = **180 m** from the stop line.

3. Journey time = 180 ÷ 13.3 = **13.5 s**: inside the 8–15 s zone.

4. Three lanes, at most two per loop: **2 loops** (lanes 1–2, lane 3).

Link N3A gets two 2 m loops at 180 m (13.5 s). Over 30 links (6 of them three-lane) Riverside needs 24 + 12 = 36 loops.



Remember it: Detectors



PICTURE THIS · STOP 4: GOLD HOOPS IN THE ROAD

Gold hula hoops lie in the road 13 big strides before the traffic lights. Each hoop is just wide enough for two lanes, and a lookout sits in every hoop shouting 'coming!' at each car.

TRAP

'Upstream' does not mean 'as far as possible'. The time from loop to stop line must be consistent: a loop before a bus stop or a car park entrance gives the model the wrong arrival times.

5 Wire it up and build the database

IN PLAIN WORDS

Wiring SCOOT is like setting up a nervous system: the loops are nerve endings, the cables are nerves, and the UTC computer is the brain. The database is the brain's memory of what each road is like.

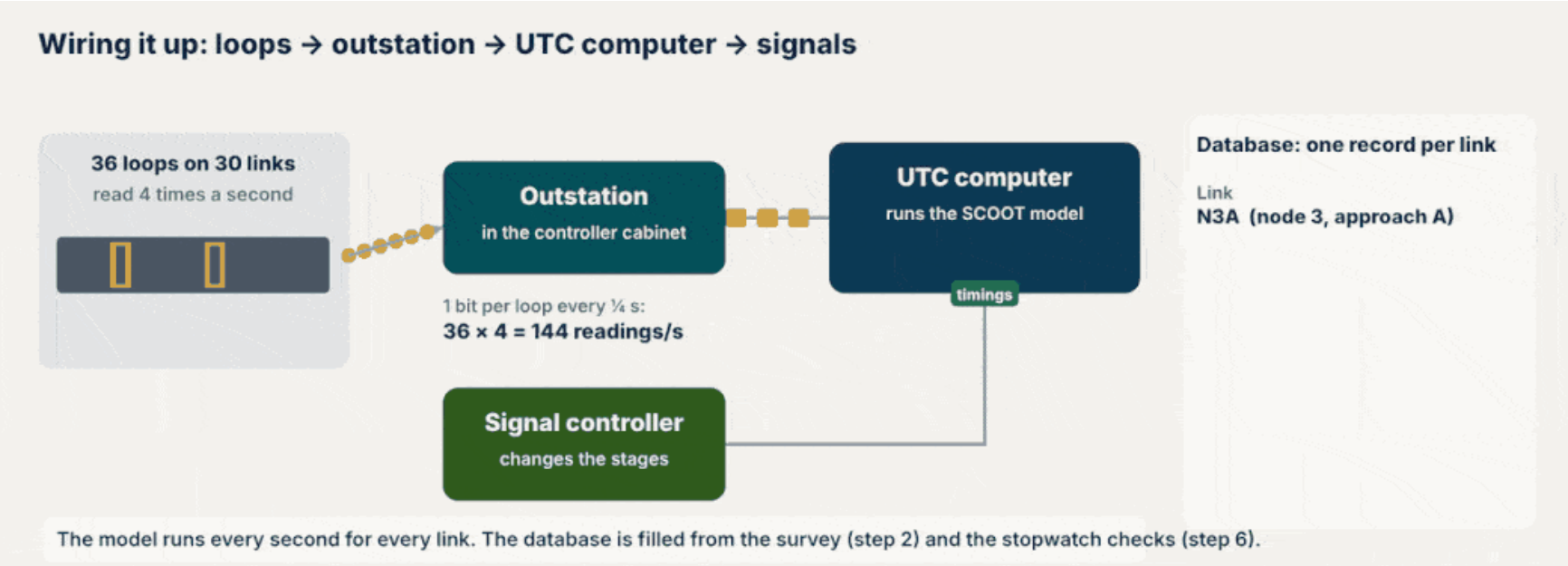
- Loops → outstation → UTC computer → signals
- Each loop read about 4 times a second
- The database describes every link and stage
- Fault monitoring watches loops and comms

144/s

loop readings for Riverside's 36 loops

Source: Central computer, communications and fault monitoring: DfT Traffic Advisory Leaflet 7/99; UTC and detection: DfT Traffic Signs Manual Ch. 6 (2019), sections 9.5 and 10.

Watch: Loops to computer and back



NOTICE 1 Readings travel up from the street several times a second.

NOTICE 2 Timings travel back down to the signal controller.

NOTICE 3 The database card fills field by field; SATO is set later, in step 6.

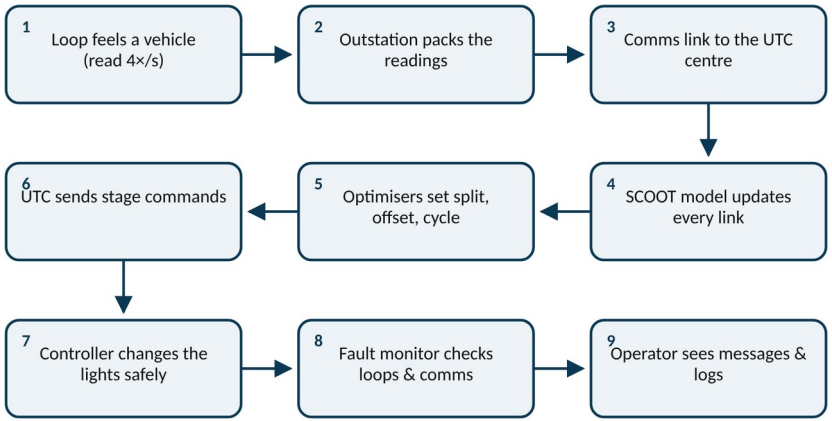
Worked example: Filling link N3A's record

readings per second = loops × 4
max queue clear time ≈ (queue length ÷ space per vehicle) ÷ s + start lag

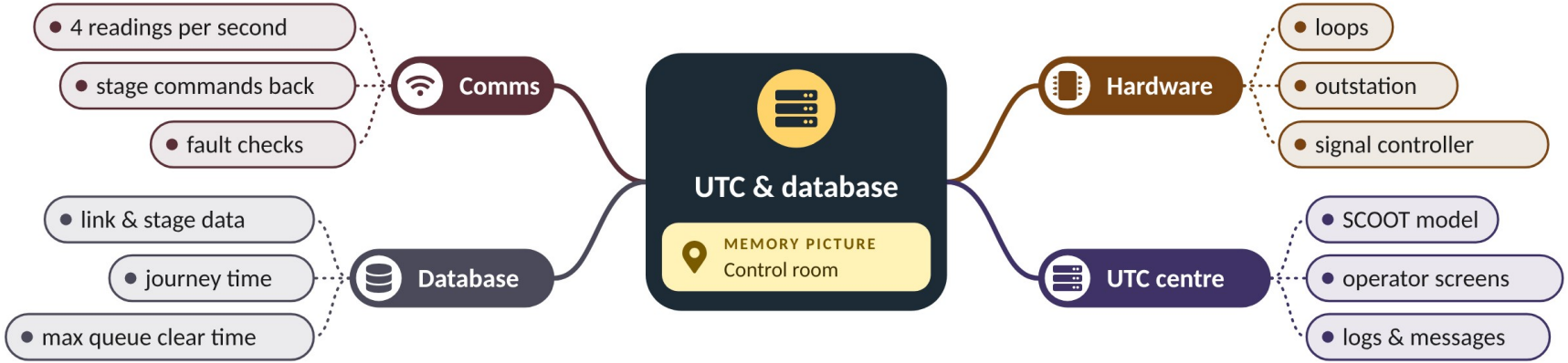
36 loops in Riverside. Link N3A: loop 180 m from the stop line, 6 m of road per queued vehicle, s = 0.5 veh/s per lane, start lag 2 s. Three stopwatch runs from loop to stop line: 13.2, 13.8 and 13.5 s.

1. Data rate: 36 × 4 = **144 readings per second**.
2. Journey time = (13.2 + 13.8 + 13.5) ÷ 3 = **13.5 s**.
3. A full queue reaches the loop: 180 ÷ 6 = **30 vehicles** per lane.
4. Max queue clear ≈ 30 ÷ 0.5 + 2 = **62 s**.

N3A's record: journey time 13.5 s, max queue clear time 62 s. Step 6 checks both on street.



Remember it: UTC & database



PICTURE THIS · STOP 5: CONTROL ROOM

In the control room a hungry computer gobbles 144 jelly beans a second through a long straw from the street. Beside it a librarian files a recipe card for every road: '13.5 s to the stop line, 62 s to clear the queue'.

TRAP

The controller in the street still enforces safety: minimum greens and intergreens. SCOOT asks for stage changes; it can never order an unsafe one.

6 Validate every link on the street

IN PLAIN WORDS

Validation is like tuning a guitar by ear. You pluck the string (watch a real queue clear), compare with the note you expect (the model's queue), and turn the peg (SATO) until they match.

- Calibrate the model to the street, link by link
- Check journey time, max queue clear time and lags
- Tune SATO until model and street queues clear together
- Use normal days and several cycles

0.59

LPU/s: N3A's validated SATO

Source: Validation = calibrating the model to the street: DfT Traffic Advisory Leaflet 7/99; journey time and max queue clear time as key parameters: Davies; procedures in the SCOOT Traffic Handbook (TRID 457437).

Watch: Validating SATO with a stopwatch



NOTICE 1 Grey bar = the real queue; blue bar = SCOOT's model queue.

NOTICE 2 With SATO 0.67 the model clears 4 s early.

NOTICE 3 Turning SATO down to 0.59 makes the two clear together.

Worked example: Correcting SATO on link N3A

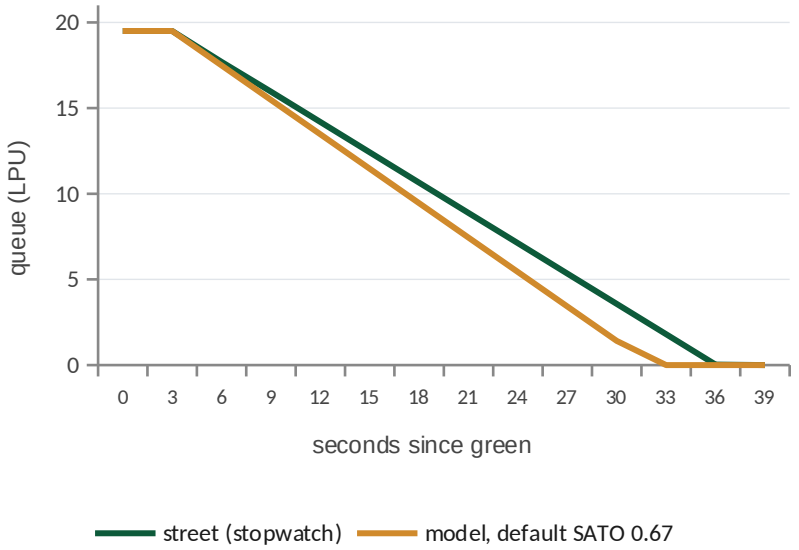
SATO = queue (LPU) ÷ (observed clear time − start lag)

Queue of 10 cars ≈ 19.5 LPU. Start lag 3 s. Default SATO 0.67 LPU/s. Stopwatch: the real queue clears 36 s after green starts.

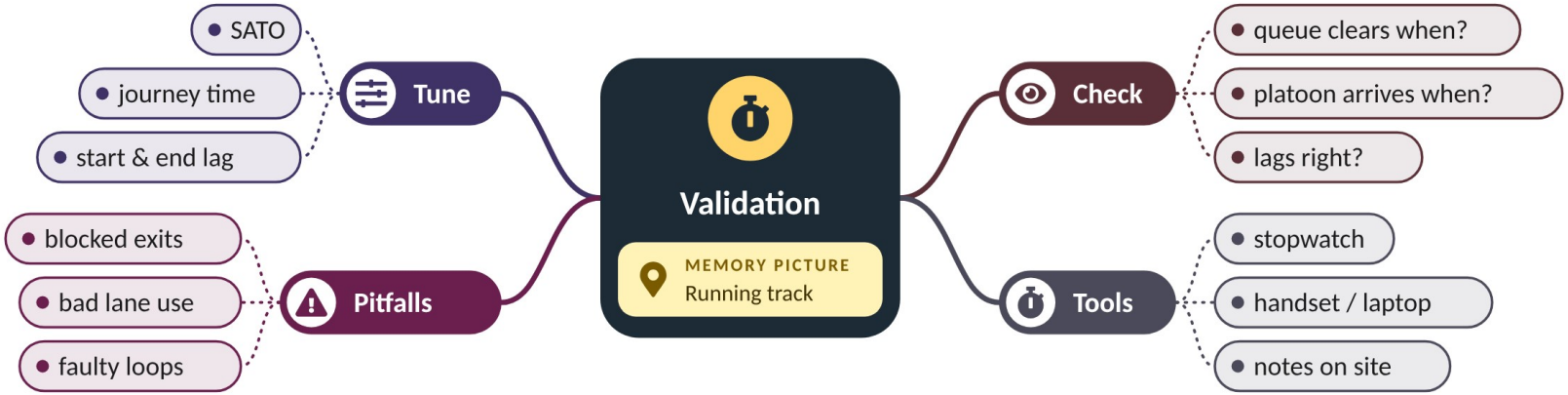
1. Model clears at 3 + 19.5 ÷ 0.67 = **32.1 s**.
2. Street clears at **36 s**: the model is about 4 s early, so SATO is too high.
3. New SATO = 19.5 ÷ (36 − 3) = 19.5 ÷ 33 = **0.59 LPU/s**.
4. Check: 3 + 19.5 ÷ 0.59 = **36.1 s**, matching the street.

SATO for N3A is set to 0.59. Repeat over several cycles and at different times of day before accepting it.

Queue clearing on N3A: street vs model before validation.



Remember it: Validation



PICTURE THIS · STOP 6: RUNNING TRACK

On a running track, engineers with stopwatches race a robot queue. The robot always finishes 4 seconds early, until an engineer turns the dial on its back from 0.67 down to 0.59. Now they cross the line together.

TRAP

Never validate on a day with road works, an incident or a blocked exit. You would tune the model to an unusual day and it would be wrong on every normal day.

7 Commission, fine-tune and evaluate

IN PLAIN WORDS

Commissioning is like the first weeks in a new house. You move in (switch on), fix the doors that stick (fine-tune), and check the heating bill against the old house (after study).

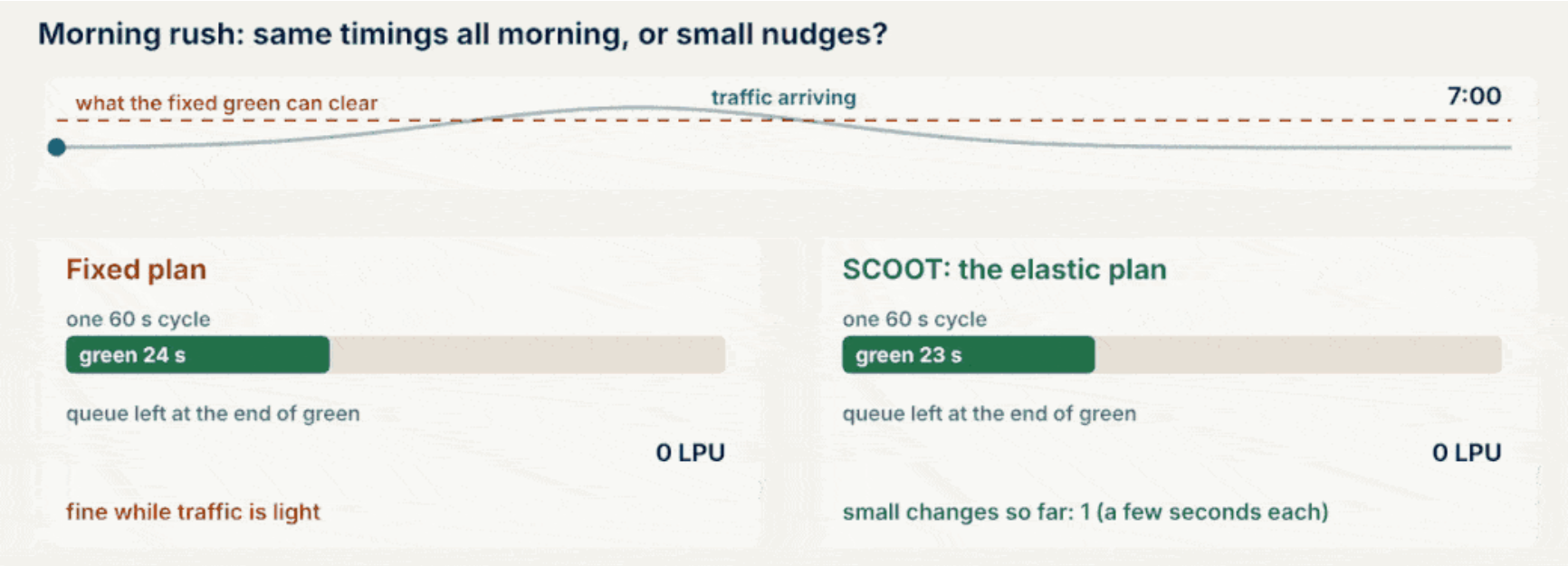
- Switch on region by region, fallback ready
- Fine-tune: gating, bus priority, congestion links
- After study on the same routes and times
- Keep checking loops and re-validating

13–15%

shorter peak journeys in Riverside (illustrative)

Source: Gating: DfT TAL 7/00; bus priority: DfT TAL 8/00; congestion and importance factors, ASTRID/INGRID: DfT Traffic Advisory Leaflet 7/99.

Watch: The elastic plan after switch-on



NOTICE 1 The timings now stretch with demand instead of staying fixed.

NOTICE 2 Changes are small and frequent, so the network never jumps.

NOTICE 3 This is what the after study measures.

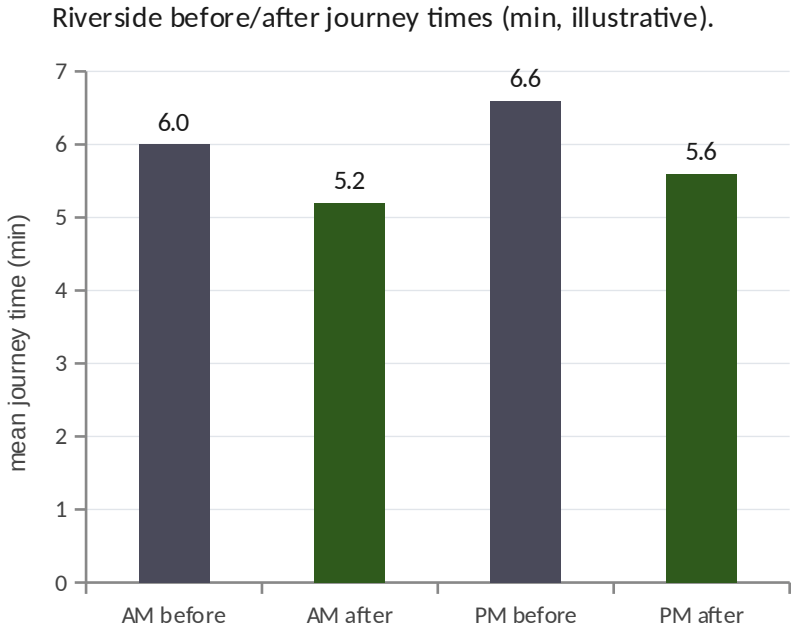
Worked example: Riverside's after study

journey-time saving % = (before − after) ÷ before × 100

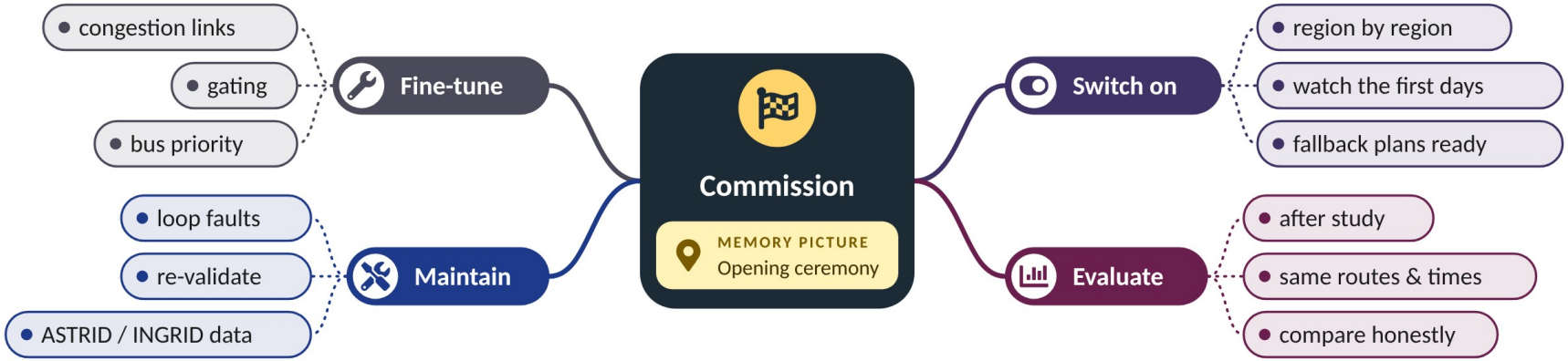
Same corridor, same weekdays, 12 runs each. Mean journey time AM peak: 6.0 min before, 5.2 min after. PM peak: 6.6 min before, 5.6 min after.

1. AM saving = (6.0 − 5.2) ÷ 6.0 = 0.8 ÷ 6.0 = **13.3%**.
2. PM saving = (6.6 − 5.6) ÷ 6.6 = 1.0 ÷ 6.6 = **15.2%**.
3. Both sit within the 12–20% range in TAL 7/99.
4. Report both, with the number of runs, not just the better one.

Riverside's journeys are about 13–15% shorter at the peaks: the feasibility estimate in step 1 was reasonable.



Remember it: Commission



PICTURE THIS · STOP 7: OPENING CEREMONY

At the opening ceremony giant scissors cut a ribbon of traffic lights. A scoreboard flips from 6.0 to 5.2, and a mechanic in overalls keeps tightening bolts on the lights long after the crowd has gone home.

TRAP

An after study on a quiet August day compared with a before study in busy November proves nothing. Match the season, weekday and time, and count enough runs.

Practical questions

Fourteen questions engineers ask: detectors, siting, scheme size, data, standards, sources, mistakes, stages, cycle limits, failures, congestion, alternatives, team, report.

Which kind of detector, and why?

SCOOT needs to know, several times a second, whether a vehicle is over one spot on each link. Any detector that can do that reliably will work; loops are the reference.

Detector	How it senses	Why use it for SCOOT	Watch out for
Inductive loop (cut into the road)	a vehicle's metal changes the loop's inductance	SCOOT was designed around loops: steady presence and occupancy at a fixed spot	lane closures to cut and repair; broken by road works
Magnetometer (core-drilled stud)	change in the Earth's magnetic field	wireless and quick to fit; adopted at over 1,000 TfL junctions, including SCOOT	battery life; must give loop-like occupancy
Radar / microwave (above ground)	reflected radio waves	no road cutting; TRL has tested radar for SCOOT	motion-only types miss stopped vehicles (TAL 16/99)
Video camera	image processing of drawn zones	many lanes from one pole; can also count turns	accuracy drops at night, in rain and low sun
Bus detection (AVL, transponder)	the bus reports where it is	needed for bus priority (TAL 8/00)	only equipped buses are seen

Sources: TRL PPR087; TAL 16/99; JCT Symposium 2019 (magnetometers); TTI report 0-5845-1; TAL 8/00.

Where exactly does each detector go?

The rule is the same everywhere: give the model warning, and keep the time from detector to stop line consistent. Special layouts need special answers.

Normal link

At the upstream end, 8–15 s before the stop line (never under about 6 s), often just past the upstream junction's exit.

Short link

If the junction upstream is close, put the loop on its exit: the warning is short but still useful.

Stop-line detector

Where upstream siting is impossible: 0–20 m from the stop line, with the link journey time set to zero.

Poor lane discipline

With very poor lane use or a short flare, the loop can go downstream of the junction instead.

Bus detector

Far enough upstream that SCOOT can extend or recall the green before the bus arrives (TAL 8/00).

Avoid

Bus stops, parking and loading bays, side-road entries and lane changes between loop and stop line; keep cable runs short.

Sources: Davies (SCOOT design principles); TAL 7/99; TAL 8/00; FDOT (2016).

How many signals should the scheme include, and how do I choose?

Start from the problem, then follow the traffic: include every signal whose platoons reach the next one.

Start with the problem

Use the before study: which routes and junctions have the worst delay and the most variable traffic?

Follow the platoons

Include neighbouring signals close enough for platoons to matter. Junctions about 1 km or more apart are often better on MOVA (TAL 7/99).

Count the crossings

Pedestrian crossings on the routes become SCOOT nodes too; many can double-cycle.

Group into regions

Clusters that should share a cycle form regions (step 3); long links make the boundaries.

Check the hardware

Each signal needs a suitable controller and a communication link to the UTC centre.

Riverside's answer

8 junctions + 2 crossings = 10 nodes, 2 regions, 30 links, 36 loops.

Sources: TAL 7/99; FHWA-HOP-11-027 (define the scope in a concept of operations).

What traffic data should be collected?

Collect what the model needs (step 5), what validation checks (step 6) and what the before and after studies compare (steps 1 and 7).

Data	How	Used for	Step
Turning counts, 15-min, peak and off-peak	manual or video counts	flow ratios Y, regions, stages	2–3
Saturation flows	stopwatch at the stop line	capacity; starting value for SATO	2, 6
Link journey (cruise) times	stopwatch or GPS runs	journey time in the database; offsets	2, 5
Queue lengths and clear times	watch the back of the queue	max queue clear time; validation	5–6
Route journey times, before and after	floating-car runs, same days and times	the benefit of the scheme	1, 7
Bus routes, stops and frequencies	operators' timetables, site visits	bus priority; loop siting	4, 7
Pedestrian demand at crossings	counts, push-button use	double cycling; pedestrian stages	3
Lane use, parking, loading	site visits, photos	where loops can and cannot go	4

Sources: TRID 186642 (practical implementation of SCOOT); TAL 7/99; Traffic Signs Manual Ch. 6.

Which references and standards, and why?

Use UK SCOOT guidance for how SCOOT works, a systems-engineering guide for planning the project, and your own national rules for what is legal and safe.

Document	Why use it
DfT TAL 7/99: The SCOOT UTC system	the official overview: regions, detectors, optimisers, validation, benefits
DfT TAL 7/00 and 8/00	how to design gating and bus priority
DfT TAL 16/99 (and spec. TR 2123)	rules for above-ground detectors
Traffic Signs Manual Ch. 6 (2019)	UK practice for signal junctions, UTC and detection
SCOOT Traffic Handbook (TRL, with the licence)	the detailed database and validation procedures
FHWA-HOP-11-027 (2012)	templates: concept of operations, requirements, verification and validation plans
NCHRP Synthesis 403 (2010)	lessons from agencies that deployed adaptive control
Your national / city signal manual	local law and safety rules (intergreens, pedestrian times) come first

Full details in the reference list at the end of the deck.

Simple sources to start implementing

You can learn and test most of the design before touching a real road.

Free UK leaflets

DfT Traffic Advisory Leaflets (7/99, 7/00, 8/00, 16/99) are free on gov.uk.

Traffic Signs Manual Ch. 6

Free from gov.uk; read sections 9.5 (UTC) and 10 (Detection).

FHWA ASCT model documents

Fill-in templates for scoping and specifying an adaptive signal project.

SUMO simulation

Free, open-source traffic simulator; this lab's SUMO labs F1–F3 test fixed, actuated and SCOOT-style control.

This lab's animations

The design GIFs and the junction simulation in this deck, on the course website.

Supplier training

Real SCOOT databases are built with TRL's licensed tools and handbook; plan training into the budget.

SUMO: eclipse.dev/sumo · Lab: smrahmanbd.github.io/scoot-learning-lab

Common design mistakes (and how to avoid them)

Most failed adaptive schemes did not fail because of the software.

Skipping validation

An unvalidated model times the lights for an imaginary town. Budget days on street for every link.

Loops in the wrong place

Too close, before a bus stop, or across three lanes. Site them by seconds and check on street.

No maintenance plan

Many adaptive systems were switched off early for lack of staff and money (FHWA-HOP-11-027).

Unfair before/after

Different season or weekday makes the result meaningless. Match days, times and routes.

Forgetting pedestrians

Crossings are nodes too, and pedestrian times are safety rules, not options.

One giant region

Forcing far-apart clusters onto one cycle makes the quiet ones wait. Split at long links.

Sources: FHWA-HOP-11-027; TAL 7/99; Stevanovic (2010) NCHRP Synthesis 403.

Stages, phases and the controller

SCOOT decides when to change stage; the street controller decides what is safe. Design both together.

Define the stages

List each node's stages and their order, and which links feed each stage. This goes into the database.

Safety stays local

Minimum greens and intergreens are set in the controller; SCOOT cannot shorten them.

Pedestrian stages

Size crossing times from your local manual's walking speed (TSM Ch. 6 in the UK); demand-dependent stages run only when called.

Min and max greens

Give every stage a minimum (safety, pedestrians) and a sensible maximum, so no movement waits forever.

Fallback plans

Every controller keeps a local plan (fixed time or VA) to run if it loses contact with the centre.

Check on site

Before switch-on, test that each stage request from the centre gives the expected lights.

Sources: TAL 7/99; DfT Traffic Signs Manual Ch. 6 (2019).

Cycle limits, pedestrians and cyclists

The cycle optimiser works inside limits that you set. Those limits are where people on foot and on bikes are protected.

Minimum cycle

Long enough for every stage's minimum green, including the pedestrian stages.

Maximum cycle

Short enough that pedestrians and side roads do not wait too long, and queues fit in the road space.

Double cycling

Crossings and quiet nodes can run twice per region cycle, halving pedestrian waits (TAL 7/99).

Walking speed

Use the design walking speed from your own standard (UK: TSM Ch. 6) for crossing times.

Cycle detection

Bicycles can be detected (for example by in-ground radar) where they need their own stage.

Newer SCOOT

SCOOT 7 is aimed at multi-modal networks: cycling, walking, public transport (TRL Software).

Sources: TAL 7/99; Traffic Signs Manual Ch. 6; TRL Software, SCOOT version history; JCT Symposium 2019.

Communications, and what happens when things fail

SCOOT is centralised, so the design must say what happens when a loop, a cable or the computer fails.

Failure	What happens	Design for it
Loop stops working	fault monitoring flags the loop; that link's model loses its input	fault reports checked daily; repair contracts with fast response
Loop damaged by road works	wrong or missing counts until repaired	agree a process with street works teams; re-validate after
Comms link lost	the controller cannot hear the centre	each controller holds a local fallback plan
Central computer down	every node loses SCOOT control	standby computer or server; fallback plans everywhere
Road layout changed	the database no longer describes the street	update the database and re-validate the links

Sources: TAL 7/99 (fault monitoring, communications, central processor).

Congestion, incidents and special features

SCOOT works best below saturation. For the busy edges of the day it has extra tools; design which ones you need.

Oversaturation

When demand exceeds capacity every adaptive system struggles (FDOT 2016). Decide where the queues should be.

Gating

Hold traffic back at chosen links to protect a sensitive area or a bus route (TAL 7/00).

Bus priority

Extend or recall green for detected buses (TAL 8/00).

Congestion settings

Congestion offsets and importance factors favour links that block back (TAL 7/99).

ASTRID and INGRID

ASTRID stores SCOOT's data for reports; INGRID helps detect incidents (TAL 7/99).

Emissions

SCOOT can estimate vehicle emissions from its model (TAL 9/00).

Sources: TAL 7/99, 7/00, 8/00, 9/00; FDOT (2016).

SCOOT or something else?

SCOOT is one option. A good design report says why it beats the alternatives for this network.

Option	Best when	Less good when
Fixed-time plans (UTC)	traffic is regular and plans are kept up to date	traffic varies; plans age about 3% a year
VA or MOVA	isolated junctions, about 1 km or more apart (TAL 7/99)	closely spaced junctions that need coordinating
SCATS	stop-line detection is preferred; distributed control	you want the link model and upstream warning
SCOOT	close junctions, variable traffic, reliable comms	heavy oversaturation; no budget for upkeep

Sources: TAL 7/99; FDOT (2016); this lab's SCATS course.

Who does what, and in what order?

A SCOOT scheme is a team project. FHWA advises writing a concept of operations first, so everyone agrees what the system is for.

Phase	Main tasks	People
Plan (step 1)	before study, business case, concept of operations	client, traffic planner
Design (steps 2–5)	survey, regions, loop drawings, database	traffic signal engineer
Build	cut loops, cables, outstations, comms	contractor, signal technicians
Validate (step 6)	stopwatch checks link by link	engineer + technician on site
Operate (step 7)	switch-on, after study, faults, re-validation	UTC operators, maintainers

Sources: FHWA-HOP-11-027 (2012); TAL 7/99.

What goes in the design report?

If another engineer can rebuild your design from the report, it is complete.

Network inventory

Every node: lanes, stages, intergreens, controller, cabinet location.

Region diagram

Regions, critical nodes, double-cycled nodes, links and their numbers.

Loop drawings

Each loop's position, size and lanes, with its journey time to the stop line.

Database sheets

Link and stage data: journey times, max queue clear times, lags, SATO.

Validation records

Who, when, what was measured, what was changed on each link.

Before/after report

Routes, days, number of runs, results and honest limitations.

Sources: TRID 186642; TAL 7/99; FHWA-HOP-11-027.

Simulation: one busy junction, old plan vs SCOOT-style plan

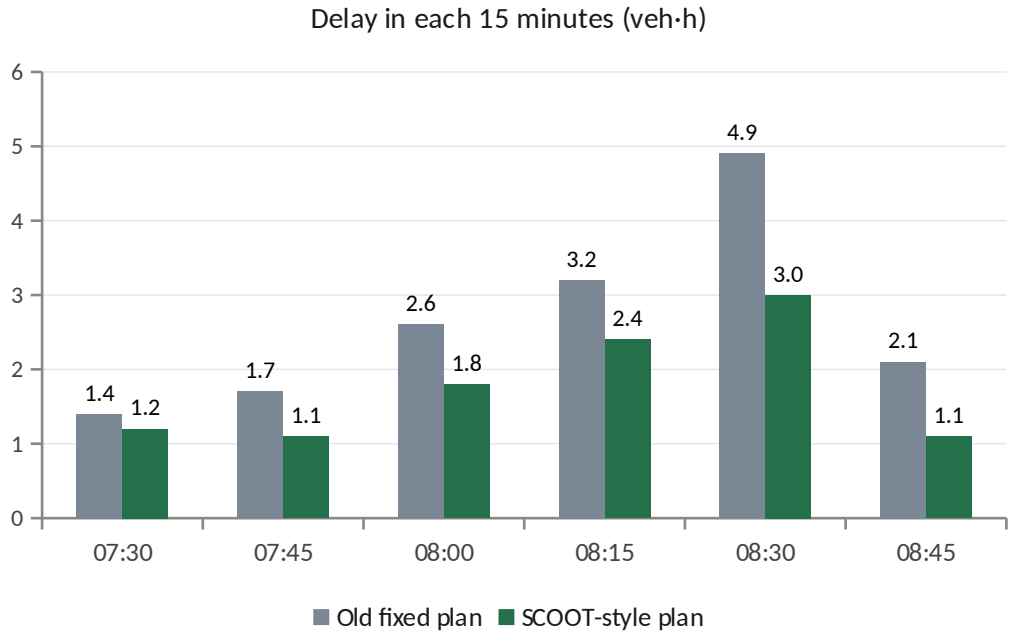


NOTICE 1 Demand on the main road peaks at 08:15; a side-road surge follows at 08:35.

NOTICE 2 The old 70 s plan cannot change, so queues build at both peaks.

NOTICE 3 The elastic plan shortens the cycle when it is quiet and shifts green to the side road during its surge.

What the simulation shows



32% less delay

15.8 → 10.7 veh·h · 20.2 → 13.6 s per vehicle

› One junction, two stages, 90 minutes, simulated second by second (a fluid queue model).

› Old plan: 70 s cycle, green 38 s main : 22 s side, never changed.

› Elastic plan: splits move up to 4 s per cycle; the cycle is reviewed every 5 min (2.5 min while rising) towards 90% saturation.

› Result: 15.8 → 10.7 veh·h of delay; mean 20.2 → 13.6 s per vehicle.

HONEST NOTE A simplified teaching model of SCOOT's ideas, not TRL's SCOOT. One isolated junction with a surge gains more than a whole network usually does (about 12–20% in TAL 7/99).

The design checklist on one page

Step	You collect	You decide	You hand on
1 Feasibility	before-study delay, costs	go / no go, scheme area	baseline, budget
2 Survey	maps, counts, s, journey times	what each node is	network inventory
3 Regions	node flows (Y), link lengths	regions, critical nodes, double cycling	region & node list
4 Detectors	lanes, speeds, kerb features	link list, loop positions	loop drawings
5 UTC & data	cable routes, controller types	comms, database values	first-guess database
6 Validation	stopwatch times on street	SATO, lags, journey times	validated model
7 Commission	after-study journey times	tuning features, maintenance	results report

Things beginners mix up

This…	…is not this	The fact that tells them apart
Region	**Node**	A node is one junction or crossing; a region is a group of nodes sharing one cycle.
Saturation flow s	**SATO**	s is vehicles per hour of green; SATO is SCOOT's version in LPU per second, set by validation.
Distance	**Journey time**	Loops are placed by seconds (8–15 s); metres depend on the street's speed.
Commissioning	**Validation**	Validation tunes each link's model; commissioning switches the system on and checks it.
Critical node	**Biggest junction**	It is the node needing the longest cycle (highest Y), not the one with most lanes.

Check yourself (1 of 2)

Two nodes need 60 s and 90 s cycles, 900 m apart. What is sensible?

A. One region at 60 s

B. One region at 90 s

C. Two regions, split at the long link

D. Turn SCOOT off at one node

At 48 km/h (13.3 m/s), which loop position is in the ideal zone?

A. 30 m from the stop line

B. 70 m

C. 160 m

D. At the stop line

Answers are in the speaker notes.

Check yourself (2 of 2)

A 3-lane approach needs how many SCOOT loops?

A. 1

B. 2

C. 3

D. 6

Before 7.5 min, after 6.6 min on the same route and time. The saving is:

A. 9%

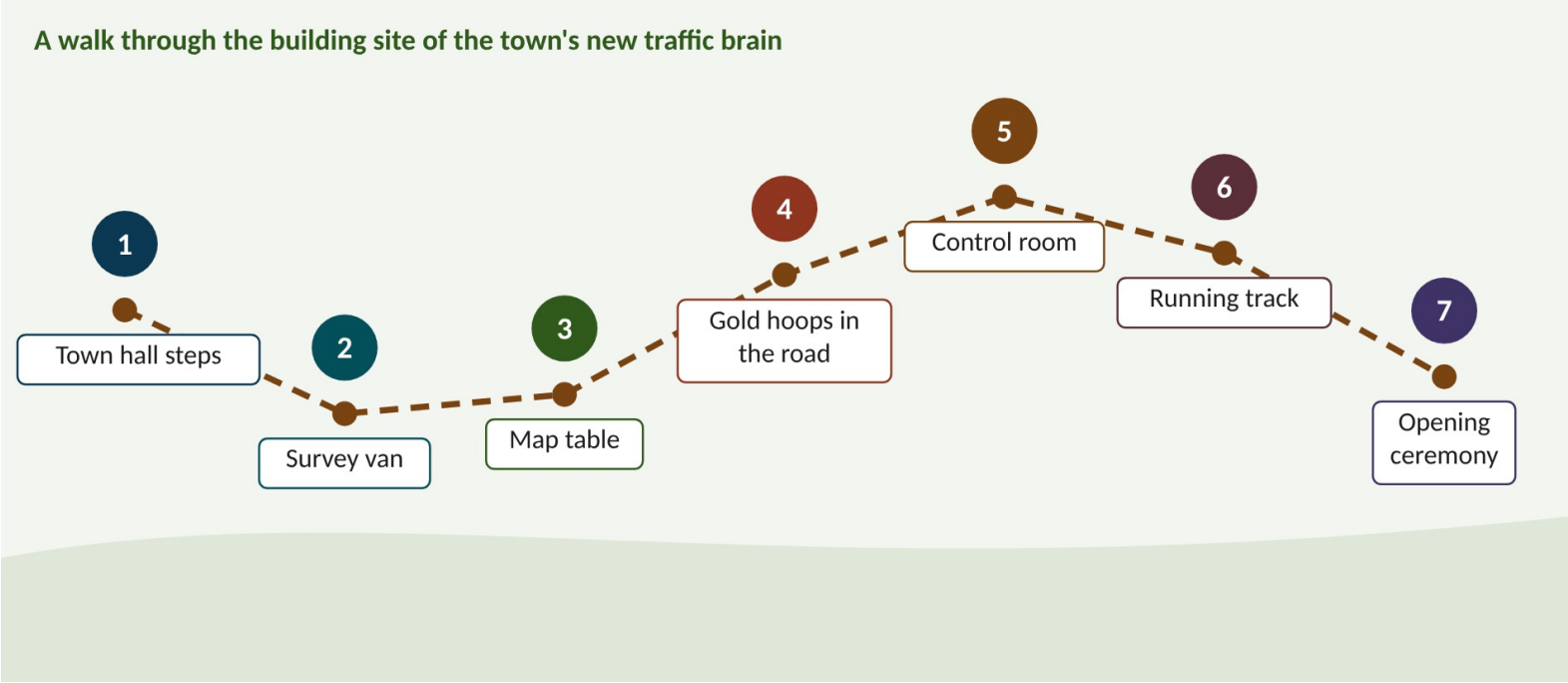
B. 12%

C. 14%

D. 0.9%

Answers are in the speaker notes.

Walk the memory route



1 Town hall steps = Feasibility · 2 Survey van = Survey · 3 Map table = Regions · 4 Gold hoops in the road = Detectors · 5 Control room = UTC & database · 6 Running track = Validation · 7 Opening ceremony = Commission

Numbers to remember

- Savings ~12% vs fresh plans, ~20% vs typical (TAL 7/99)
- Plans age ~3% a year
- s ≈ 1800 veh/h per lane; count from vehicle 1
- Critical node target ≈ 90%; C = 0.9L ÷ (0.9 − Y)
- Loops 8–15 s upstream (≥ 6 s), 2 m, ≤ 2 lanes
- 48 km/h = 13.3 m/s: 8 s = 107 m, 15 s = 200 m
- Each loop read 4 times a second
- SATO = queue ÷ (clear time − start lag)

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21. SCOOT Learning Lab: Foundation, Advanced and Mastery pages, SUMO labs. smrahmanbd.github.io/scoot-learning-lab. [all]

Keep going

Next: SCOOT Foundation deck (how the model works) · Design animations and simulation: smrahmanbd.github.io/scoot-learning-lab